

Physics Assignment

CLASSMATE

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Simple Harmonic Motion (SHM)

1. (a) Since $x = A \cos(\omega t + \phi)$, we see that $A =$

$$A = 4.00 \text{ m} \quad \& \quad \omega = \pi \text{ rad/s. Therefore,}$$

$$f = \frac{\omega}{2\pi} = \frac{\pi}{2\pi} = \frac{1}{2} = 0.500 \text{ Hz}$$

$$\text{and } T = \frac{1}{f} = \frac{1}{1/2} = 2 \text{ s}$$

(b) Differentiating x to find v to find a , we obtain

$$v = \frac{dx}{dt} = -(4.00 \pi \text{ m/s}) \sin\left(\pi t + \frac{\pi}{4}\right)$$

$$\text{and } a = \frac{dv}{dt} = -(4.00 \pi^2 \text{ m/s}^2) \cos\left(\pi t + \frac{\pi}{4}\right)$$

(c) At $t = 1 \text{ s}$

$$x = (4.00 \text{ m}) \cos\left(\pi + \frac{\pi}{4}\right) = (4.00 \text{ m}) \cos\left(\frac{5\pi}{4}\right)$$

$$= 4 \times -0.707 = -2.83 \text{ m}$$

$$v = -(4.00 \pi \text{ m/s}) \sin\left(\frac{5\pi}{4}\right)$$

$$= -(4.00 \pi \text{ m/s}) (-0.707) = 8.89 \text{ m/s}$$

$$a = -(4.00 \pi^2 \text{ m/s}^2) (-0.707) = 27.9 \text{ m/s}^2$$

d) Since maximum values of sine & cosine functions are unity. Therefore, v varies between ~~$\pm 4.00/\pi$~~
 $\pm 4.00 \pi \text{ m/s}$ and a varies between
 $\pm 4.00 \pi^2 \text{ m/s}^2$. Thus,

$$v_{\max} = 4.00/\pi \text{ m/s} \text{ and} \\ = 12.6 \text{ m/s}$$

$$a_{\max} = 4.00/\pi^2 \text{ m/s}^2 = 39.5 \text{ m/s}^2$$

we obtain the same results using the relations $v_{\max} = \omega A$ and $a_{\max} = \omega^2 A$ where
 $A = 4.00 \text{ m}$ & $\omega = \pi \text{ rad/s}$

$$\begin{aligned}
 2. (a) \quad T &= 2\pi \sqrt{\frac{m}{k}} \\
 &= 2\pi \sqrt{\frac{0.200}{5.00}} = 1.257 \text{ s} \\
 &= 1.26 \text{ s}
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad V_{\max} &= \omega A \quad \omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{5}{0.2}} = \sqrt{25} \\
 &= 5 \times 0.0500 = 5 \text{ rad/s} \\
 &= 0.250 \text{ m/s} \\
 V_{\max} &= 0.250 \text{ m/s}
 \end{aligned}$$

$$\begin{aligned}
 c) \quad a_{\max} &= \omega^2 A = (5)^2 \times 0.0500 \\
 &= 25 \times 0.0500 \\
 &= 1.25 \text{ m/s}^2
 \end{aligned}$$

$$\begin{aligned}
 d) \quad x(t) &= A \cos(\omega t) = 0.05 \cos 5.00 t \text{ m} \\
 v(t) &= -A\omega \sin(\omega t) = -0.05 \times 5 \sin 5t \\
 &= -0.25 \sin 5.00 t \text{ m/s} \\
 a(t) &= -A\omega^2 \cos(\omega t) = -0.05 \times 5^2 \cos 5.00 t \\
 &= -0.05 \times 25 \cos 5.00 t \\
 &= -1.25 \cos 5.00 t \text{ m/s}^2
 \end{aligned}$$

q3.

$$x(t) = A \cos(\omega t + \phi)$$

$$A = 0.320$$

$$T = 0.900 \text{ s}$$

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{0.900} \approx 6.98 \text{ rad/s}$$

At $t=0$ $x=A=0.320 \text{ m}$ & $v=0$

$\therefore$ cosine function starts at maximum displacement,

$$x(t) = A \cos(\omega t)$$

a) time to go from $x=0.320 \text{ m}$ to $x=0.160 \text{ m}$

$$x(t) = A \cos(\omega t) = 0.160$$

$$\cos(\omega t) = \frac{0.160}{0.320} = 0.5$$

$$\omega t = \cos^{-1}(0.5) = \frac{\pi}{3}$$

$$t = \frac{\pi}{3\omega} = \frac{\pi}{3 \cdot 6.98} \approx 0.150 \text{ s}$$

So, time from 0.320 m to 0.160 m is 0.150 s

b) time to go from $x=0.160 \text{ m}$ to $x=0$

$$x(t) = A \cos(\omega t) = 0$$

$$\cos(\omega t) = 0 \Rightarrow \omega t = \frac{\pi}{2}$$

$$t = \frac{\pi}{2\omega} = \frac{\pi}{2 \cdot 6.98} \approx 0.225 \text{ s}$$

But this is from $x=A$ to $x=0$

We already know it took 0.150 s to reach $x=0.160 \text{ m}$

So the additional time from 0.160 m to 0 is

$$\Rightarrow 0.225 - 0.150 = 0.075 \text{ s}$$

So, time from 0.160 m to 0 is 0.075 s

$$V_{xt} = - (3.60 \text{ cm/s}) \sin \left((4.71 \text{ s}^{-1})t - \frac{\pi}{2} \right)$$

$$x(t) = A \cos(\omega t + \phi) \quad \text{so}$$

$$v(t) = \frac{dx}{dt} = -A\omega \sin(\omega t + \phi)$$

$$\omega = 4.71 \text{ s}^{-1}$$

$$A\omega = 3.60 \text{ cm/s}$$

$$(a) \quad T = \frac{2\pi}{\omega} = \frac{2\pi}{4.71} \approx 1.33 \text{ s}$$

$$(b) \quad A = \frac{v_{\max}}{\omega} = \frac{3.60}{4.71} \approx 0.764 = 7.64 \times 10^{-3} \text{ m}$$

$$= 7.64 \text{ mm}$$

$$(c) \quad a_{\max} = A\omega^2 = (7.64 \times 10^{-3})(4.71)^2$$

$$\approx 0.170 \text{ m/s}^2$$

$$(d) \quad \omega = \sqrt{\frac{k}{m}} \Rightarrow k = m\omega^2$$

$$= 0.5 \times (4.71)^2$$

$$\approx 11.1 \text{ N/m}$$

$f \rightarrow$ frequency $k \rightarrow$ spring constant
 $m \rightarrow$ mass attached

$$S. \quad f = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

$$f^2 = \frac{k}{4\pi^2 m}$$

Case 1 mass = m frequency = 0.83 Hz

$$f_1^2 = \frac{k}{4\pi^2 m}$$

Case 2 mass = $m + 0.680 \text{ kg}$ frequency = 0.60 Hz

$$f_2^2 = \frac{k}{4\pi^2 (m + 0.680)}$$

(To eliminate k)

$$\frac{\frac{4\pi^2}{f_1^2}}{\frac{4\pi^2}{f_2^2}} = \frac{\frac{k}{m}}{\frac{k}{m + 0.680}}$$

$$\frac{f_2^2}{f_1^2} = \frac{m + 0.680}{m}$$

(finding m) $m \frac{(0.83)^2}{(0.60)^2} = m + 0.680$

$$1.913 m - m = 0.680$$

$$0.913 m = 0.680$$

(Original mass) $m = \frac{0.680}{0.913} \approx 0.745 \text{ kg}$

6. $m = 0.175 \text{ kg}$ $k = 155 \text{ N/m}$ $x = 0.030 \text{ m}$ $v = 0.815 \text{ m/s}$

(a) $E = \frac{1}{2} k A^2 = \frac{1}{2} k x^2 + \frac{1}{2} m v^2$

$$A = \sqrt{x^2 + \frac{m}{k} v^2}$$

$$= \sqrt{(0.030)^2 + \left(\frac{0.175}{155}\right)(0.815)^2}$$

$$\approx 0.0406 \text{ m}$$

$$= 4.06 \text{ cm}$$

b) $v_{\max} = A \omega$ $\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{155}{0.175}} \approx 29.76 \text{ rad/s}$

$$= 0.0406 \times 29.76$$

$$= 1.21 \text{ m/s}$$

c) $\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{155}{0.175}} \approx 29.75 \text{ rad/s} \approx 29.8 \text{ rad/s}$

7. $M = 4.648 \text{ kg}$ $m = 7.870 \text{ g} = 7.87 \times 10^{-3} \text{ kg}$
 $k = 142.7 \text{ N/m}$ $x = 9.46 \text{ cm} = 9.46 \times 10^{-2} \text{ m}$

$$\frac{1}{2} (M+m) v^2 = \frac{1}{2} k x^2 \Rightarrow v = \sqrt{\frac{k x^2}{M+m}}$$

$$m v = (M+m) V \Rightarrow v = \frac{M+m}{m} V$$

$$M+m = 4.648 + 7.87 \times 10^{-3} = 4.65587 \text{ kg}$$

$$V = \sqrt{\frac{142.7 (0.0946)^2}{4.65587}} \approx 0.524 \text{ m/s}$$

$$v = \frac{4.65587 \times 0.524}{0.007870}$$

$$\approx 309.8 \text{ m/s}$$

$$\text{Bullet speed } v \approx 309.8 \text{ m/s}$$

8. $m = 1.6 \text{ kg}$ $k = 1.0 \times 10^3 \text{ N/m}$ $x_0 = 0.020 \text{ m}$

(a) The spring's initial P.E converts entirely to K.E at $x=0$

$$\therefore \frac{1}{2} k x_0^2 = \frac{1}{2} m v^2$$

$$\frac{1}{2} (1000) (0.020)^2 = \frac{1}{2} (1.6) v^2$$

$$0.20 = 0.8 v^2$$

$$v^2 = \frac{0.20}{0.8}$$

Speed at equilibrium $v = \sqrt{0.25} = 0.50 \text{ m/s}$

b) $f = 4.0 \text{ N}$ friction does negative work over the 0.020 m travel: $W_f = fd = 4.0 \times 0.020 = 0.080 \text{ J}$

The kinetic energy at $x=0$ is reduced accordingly.

$$\frac{1}{2} m v^2 = \frac{1}{2} k x_0^2 - W_f = 0.20 - 0.080 = 0.12 \text{ J}$$

$$\frac{1}{2} m v^2 = 0.12$$

$$\frac{1}{2} (1.6) v^2 = 0.12$$

$$v^2 = \frac{0.12}{0.8} \Rightarrow v = \sqrt{0.15} \approx 0.39 \text{ m/s}$$

Speed at equilibrium with friction
 $\Rightarrow v = 0.39 \text{ m/s}$

9. For small angles, a simple pendulum undergoes SHM with period $T = 2\pi \sqrt{\frac{L}{g}}$

The bob reaches its highest speed when it passes through equilibrium, which occurs one-quarter period after release.

$$t_{\text{to max speed}} = \frac{T}{4}$$

$$L = 0.240 \text{ m} \quad g = 9.81 \text{ m/s}^2$$

$$\text{Period} : T = 2\pi \sqrt{\frac{L}{g}} = 2\pi \sqrt{\frac{0.240}{9.81}}$$

$$= 2\pi (0.1564) \approx 0.983 \text{ s}$$

Time to highest speed

$$\Rightarrow \frac{T}{4} = \frac{0.983}{4} \approx 0.246 \text{ s}$$

$$= 0.25 \text{ s}$$

(a) from 3.50° $t = 0.25 \text{ s}$

b) from 1.75° $t = 0.25 \text{ s}$

In the small-angle regime, this time is independent of the initial angle; the difference b/w 3.50° & 1.75° is negligible

10. For small angles, angular displacement $\Rightarrow \theta(t) = \theta_{\max} \cos(\omega t)$
amplitude $\theta_{\max} = 10^\circ$

$$|\theta(t)| \leq 5^\circ \Rightarrow |\cos(\omega t)| \leq \frac{5}{10} = 0.5$$

$\therefore$ Pendulum is inside central region when $|\cos(\omega t)| \leq 0.5$

$$\therefore \cos(\omega t) \in [-0.5, +0.5]$$
$$\omega t \in \left[\frac{\pi}{3}, \frac{2\pi}{3}\right] \text{ \& } \omega t \in \left[\frac{4\pi}{3}, \frac{5\pi}{3}\right]$$

Total time spent in region

each interval length $\Rightarrow \frac{2\pi}{3} - \frac{\pi}{3} = \frac{\pi}{3}$

$$\Delta(\omega t) = \frac{2\pi}{3}$$

$$\frac{\Delta(\omega t)}{2\pi} = \frac{1}{3} \rightarrow \text{fraction of cycle}$$

$\therefore$ The pendulum spends one-third of its time period inside the region b/w -5° & $+5^\circ$
 $\Rightarrow \frac{T}{3}$